Check for updates

Received: 15 September 2021 | Accepted: 22 December 2021

DOI: 10.1111/jocd.14728

JCD Journal of Cosmetic Dermatology WILEY

ORIGINAL ARTICLE

OnabotulinumtoxinA from lines to facial reshaping: A new Italian consensus report

Massimo Signorini MD[1] | Salvatore Piero Fundarò MD[2] | Dario Bertossi MD[3] | Maurizio Cavallini MD[1] | Pierfrancesco Cirillo MD[4] | Guglielmo Natuzzi MD[5] | Sandro Quartucci MD[4] | Chantal Sciuto MD[4] | Mariagrazia Patalano MD[6] | Gloria Trocchi MD[4]

[1]Private Practice, Milan, Italy

[2]Multimed, Bologna, Italy

[3]Section of Oral and Maxillofacial Surgery, Department of Surgical Sciences, University of Verona, Verona, Italy

[4]Private Practice, Rome, Italy

[5]San Camillo Forlanini Hospital, Rome, Italy

[6]Private Practice, Messina, Italy

Correspondence
Salvatore Piero Fundarò, Multimed, Via dei Fornaciai 29/b, Bologna 40129, Italy.
Email: dr.fundaro@gmail.com

Funding information
Writing and editorial assistance was provided to the authors by Timothy Ryder, DPhil, of Biological Communications Limited (London, United Kingdom) and funded by Allergan SpA at the request of the investigator. All authors met the ICMJE authorship criteria. Neither honoraria nor payments were made for authorship

Abstract

Background: Botulinum neurotoxin type A is the most widely used nonsurgical treatment for esthetic improvement of the face and neck. In 2015, an Italian consensus panel met to discuss the optimal methods for using onabotulinumtoxinA. However, clinical practice continues to evolve and the original report is now considered obsolete.

Aims: To provide updated guidance on the esthetic uses of onabotulinumtoxinA in the face and neck.

Methods: A panel of 10 Italian specialists (including plastic and maxillofacial surgeons, dermatologists, and esthetic doctors) individually completed a questionnaire on their own clinical practice, and then met to discuss their responses and agree on a revised treatment consensus.

Results: Recommendations are provided on patient assessment, reconstitution of onabotulinumtoxinA, and preferred procedures (injection sites, doses, anatomical targets, safety precautions, etc.) across a variety of treatment areas, including glabellar, crow's feet, and forehead lines; brow lifting and shaping; lower eyelid hypertrophy; bunny lines; sagging nasal tip; gummy smile; masseter hypertrophy; perioral lines; marionette lines and "sad mouth;" mentalis hypertonia; and platysma bands. Some of the recommended doses are substantially increased from the previous consensus (particularly in the upper third and masseter) for the purpose of achieving longer lasting results without affecting safety. Furthermore, two increasingly popular techniques—the Nefertiti lift and Microbotox—are included in the consensus for the first time.

Conclusions: Optimal practice with onabotulinumtoxinA requires a systematic approach to maximize safety and effectiveness across the range of potential uses. The present consensus was developed to support these aims.

KEYWORDS
BoNTA, botulinum neurotoxin type A, esthetic medicine, facial reshaping, onabotulinumtoxinA

Massimo Signorini and Salvatore Piero Fundarò contributed equally to this work.

1 | INTRODUCTION

Use of botulinum neurotoxin type A (BoNTA) for the esthetic treatment of facial and neck wrinkles continues to increase year on year.[1,2] Indeed, BoNTA remains the most commonly used of all the available minimally invasive esthetic techniques, accounting for almost half of the estimated 13.6 million procedures performed by plastic surgeons worldwide in 2019.[2] In Italy, BoNTA treatments accounted for more than 40% of all nonsurgical esthetic procedures over this period.[2]

With more and more practitioners using these techniques, the need for appropriate guidance becomes ever more pressing. In 2015, a panel of Italian specialists (including plastic and maxillofacial surgeons, dermatologists, and esthetic doctors) met to discuss the optimal methods for treating patients using onabotulinumtoxinA—the first BoNTA formulation approved for use in esthetic medicine and still the most widely studied. From these discussions, the panel published the first "Italian consensus report on the esthetic use of onabotulinumtoxinA."[3] Recommendations were made on patient assessment, reconstitution of onabotulinumtoxinA, and treatment procedures (including injection sites and doses) for the following areas: glabellar lines, crow's feet lines, forehead lines, brow lifting and shaping, lower eyelid hypertrophy, bunny lines, sagging nasal tip, gummy smile, masseter hypertrophy, perioral lines, marionette lines, mentalis hypertonia, and platysma bands. This consensus was translated into English (with some minor updates) in 2018.[4]

However, a great deal has changed in recent years. In particular, the license for onabotulinumtoxinA has been updated to include the treatment of forehead lines. Hence, it is currently indicated for the temporary improvement of moderate-to-severe glabellar lines, lateral canthal lines (crow's feet), and forehead lines, when their severity has an important psychological impact in adult patients.[5] Furthermore, a 100-unit vial is now available both in Italy and elsewhere. New safety and efficacy findings have also led many practitioners to review the doses that they use; for example, higher doses of onabotulinumtoxinA injected in the glabella have been shown to yield clinically significant increases in the duration of effect without any detrimental impact on safety.[6] In addition, novel techniques with BoNTA products—such as the "Nefertiti lift" and "Microbotox"—have gained further traction, as has the concept of "full face treatment" with BoNTA.[7]

Thus, the original consensus report is now considered to be obsolete. To update its content, the same group of practitioners convened to reconsider their previous recommendations on the esthetic use of onabotulinumtoxinA in the face and neck—and where necessary to revise their guidance.

2 | MATERIAL AND METHODS

As per our original consensus, each of the panel of 10 Italian specialists first completed a questionnaire on their own protocols for treating the face and neck with onabotulinumtoxinA. Specifically, they were asked to describe the diluent and reconstitution of onabotulinumtoxinA, the injection sites used in each area of the face and neck, preferred doses at each site (in Allergan units [U]), and the total quantities used. Individual variations based on clinical experience could be noted in free-text form within the questionnaire.

In a second stage, the panel met to discuss the questionnaire results and agree on a treatment consensus. A "recommendation" was defined as any statement obtaining the agreement of at least 80% of the group (≥8 of 10 panel members). In instances where there was agreement of ≥60% and <80% (ie, 6–7 of the 10 panel members), the group accepted this as a"position statement."

Figures illustrating the consensus are provided throughout this report. These have been designed with innovative codes to safely guide users on preferred injection sites and needle penetration (Table 1). Each figure shows the preferred needle angle, direction and insertion depth, dosage per injection site, and the target muscle/anatomical layer.

3 | RESULTS

3.1 | Patient assessment

When evaluating a patient for esthetic treatment using onabotulinumtoxinA, a number of parameters should be assessed. General considerations include sex, age, ethnicity, skin type and scarring, history of relevant surgical and nonsurgical procedures, and any tattoos (particularly around the eyebrows or on the lips). Various specific

TABLE 1 Representations used in figures

Property	Representation in figures
Dosage (units) per injection site	• Stipulated next to each injection point
Injection depth	• Described in figure insets and color-coded accordingly: o Tip only, purple o One third needle insertion, yellow o Half needle insertion, blue o Full needle insertion, green • Target anatomical layer is shown in figure insets
Injection angle	• Stated in figure insets • Shown pictorially in each figure
Needle insertion direction	• Shown pictorially in each figure and in figure insets

parameters should also be assessed: facial shape and proportions (eg, forehead height, inter-eyebrow distance); facial asymmetries (vertical or horizontal); muscle-related tics or spastic irregularities; skin discolorations; eyelid problems (eg, upper lid ptosis, skin laxity of the lower lid, chronic edema); brow ptosis and frontalis hypercompensation; orbicularis oculi contraction (for lateral canthal lines); smile dynamics (muscle hyperactivity, exposure of the teeth and gums, and extent of occlusion); mentalis muscle hyperactivity (golf-ball or peau d'orange appearance); masseter hypertrophy and bruxism; and platysma bands.

The patient should be seated during these assessments, with specific parameters evaluated both at rest and during dynamic activity. Alterations in the position of the patient (eg, head extension or flexion) can substantially alter key assessments, potentially leading to an erroneous overall evaluation and hence to a suboptimal result.[8]

With male patients, it is important to consider anatomical differences relative to females. These include a larger overall skull size, a higher and broader forehead, more prominent supraorbital ridges, lower eyebrows with a flattened profile, a wider and more prominent glabella, squared orbits, and a more projecting and angular chin.[9-11] There are also important muscular differences. For example, men often have substantially more overall facial muscle and greater functional activity, which may predict the appearance of deeper lines over time.[11] Thus, in our experience, the treatment of specific areas of the upper third of the face may require increases in onabotulinumtoxinA dose of up to 50%–100% in males compared with females. By contrast, men typically show comparatively fewer signs of aging in the lower third of the face, owing to an increased number of pilosebaceous annexes and enhanced vascularization.[12]

Photographic documentation should be undertaken for all patients before and after treatment, following the recommendations of our previous consensus[4] and others.[10,13] Furthermore, considering the dynamic effects of BoNTA treatment, patient videos may be valuable (where possible) for complete documentation of results and for scientific assessment.

3.2 | Reconstitution of onabotulinumtoxinA

Reconstitution should be performed (just before use) to a volume that yields 4 U of onabotulinumtoxinA per 0.1 ml of solution. Thus, the recommended volume is 1.25 ml for a 50 U vial, or 2.5 ml when using a 100 U vial. The typical preferred diluent is sterile, preservative-free saline (0.9% sodium chloride solution). However, depending on the manufacturer and batch, saline solution usually has a slightly acidic pH (4.5–7), and some patients experience a feeling of "burning" during injection. Preserved (bacteriostatic) saline solution containing 0.9% benzyl alcohol has a mild local anesthetic effect and can, therefore, be used as an alternative to reduce discomfort.[14,15] However, preserved saline is not available in most European countries. It has also been reported that reconstitution of onabotulinumtoxinA using saline with added lidocaine can decrease injection pain and that the addition of adrenaline could help to minimize diffusion,[16,17] but high-quality trials are currently lacking.

For reconstitution, the panel recommends using a 2 ml syringe, usually with a 21G X 1" needle. For injection, the favored devices are typically standard (Luer lock or Luer slip) 1 ml syringes with finer needles (31–32-33G X 1/2"). The use of higher-gage needles has been shown reduce the overall pain levels associated with multiple injections of the face.[18] Nonetheless, needle gage choice should be tailored to the injector's personal preferences. Special syringes are also now available that allow needle change and have a specially designed plunger for reducing dead space at the tip. To read more precisely the graduated scale, it is possible to use a 0.3–0.5 ml insulin syringe, although this has the disadvantage of a non-interchangeable needle. Delivery devices are also available and may be preferred for precision and speed of delivery.[19]

The onabotulinumtoxinA Summary of Product Characteristics (SmPC) recommends that the reconstituted solution is used immediately, although the SmPC also notes that it remains stable for 24 h at 2–8°C.[5]However, in our experience, the reconstituted solution can be kept refrigerated at 4°C for up to 6 weeks without any substantial detrimental effects—and this is supported by study data showing no loss of potency over a few weeks of refrigeration.[20]

3.3 | Treatment area recommendations

The panel reached a consensus of ≥80% for all of the treatments described below, except gummy smile, for which only a consensus of ≥60% was reached. Hence, the following discussions can be considered as "recommendations" for all treatment areas except gummy smile, which should be considered as a "position statement."

3.3.1 | Why higher doses?

Compared with our previous consensus, some of the recommended doses have been substantially increased—particularly for the upper third of the face and the masseter. The goal is to achieve a longer lasting result without increasing the risk of side effects. Although injectors have long been aware that increasing the number of units injected can prolong the effects of onabotulinumtoxinA, the scientific basis was only elucidated following the demonstration of the role of septins in stabilizing light chain activity at the level of plasma membranes.[21] A recent prospective, double-blind, randomized, controlled trial confirmed this effect in clinical practice.[6] A total of 226 patients with moderate-to-severe glabellar lines were randomized 2:2:2:2:1 to one of four doses of onabotulinumtoxinA (20, 40, 60 or 80 U) or placebo; responses rates at 24 weeks were 30.6%–38.5% with the 40–80 U doses but only 16.0% with 20 U ($p < 0.05$ for 40 or 80 U vs 20 U), and there were no dose-related safety signals.[6]

3.4 | Glabellar lines

When treating glabellar lines, 5–7 injection sites are recommended, targeting the corrugator muscles and the procerus (Figure 1). Seven points are preferred when the tail of the corrugator extends significantly laterally.

The suggested dose of onabotulinumtoxinA is 6 U for each of the five standard points (total dose: 30 U). In cases with a powerful muscle mass and previous unsuccessful treatment, the total dose may be increased up to 50 U.[5]

For injection of the procerus, the needle should be inserted superficially (one third of its length) at a 60–90 degree angle to the skin surface. This is important because if the solution is injected to the deep fascia of the procerus, it will have only a partial effect on the muscle, and may well involve the medial part of the orbicularis oculi. The same needle is used for the medial corrugator, at the same angle, but with full penetration. For lateral corrugator injections, only one third of the needle should be inserted and at a more oblique angle. Unwanted diffusion into the levator muscle can lead to eyelid ptosis,[22] although this risk can be reduced by ensuring that lateral injections of the corrugator remain superficial and that soft tissues are elevated for 1–2 min. Moreover, for injection of the corrugator supercilii muscles, the panel recommends insertion of the needle in a superolateral direction, to keep away from the danger areas. Injections should be close to the eyebrow arch to reduce toxin spread into the caudal part of the frontalis muscles—which typically leads to reduced brow support. Cases of eyelid ptosis are always short-lived and apraclonidine hydrochloride 0.5% eyedrops can be used to contract Müller's muscle, thereby reducing the overall impact.

The panel does not advise treatment of glabellar lines by injecting only the procerus and medial corrugator muscles; this promotes overaction of the tail of the corrugator, with new wrinkles appearing at the upper orbital margin. Similarly, the procerus should always be injected even in the absence of horizontal lines at the root of the nose. Treating the glabella by injecting the corrugators alone can lead to compensatory overaction of the procerus, thus generating new lines.

3.5 | Crow's feet (lateral canthal) lines

The major contributor to crow's feet lines is the lateral part of the orbicularis oculi muscle, although the zygomatic muscles may also play a role in the formation of lower lines. Treatment is usually based on three injections given laterally to the canthus. To avoid the orbit, injections should proceed in a medial-to-lateral direction, with the needle at a 30 angle and only the tip inserted (Figure 2). The total recommended dose of onabotulinumtoxinA is 36 U for both female and male patients.

In some cases, it may be advisable to inject two further sites per side located more laterally to the standard pattern—for example if the lines continue particularly far laterally or the orbicularis oculi muscle is stronger than usual. The recommended dose is 2 U of onabotulinumtoxinA per site, thus increasing the total dose to up to 44 U (in both females and males).

The subcutaneous tissue in this area is very thin and the highly vascularized orbicularis muscle is situated almost subdermally. The panel recommends superficial injection into the deep dermis with the toxin only reaching the muscle by continuity. This decreases the

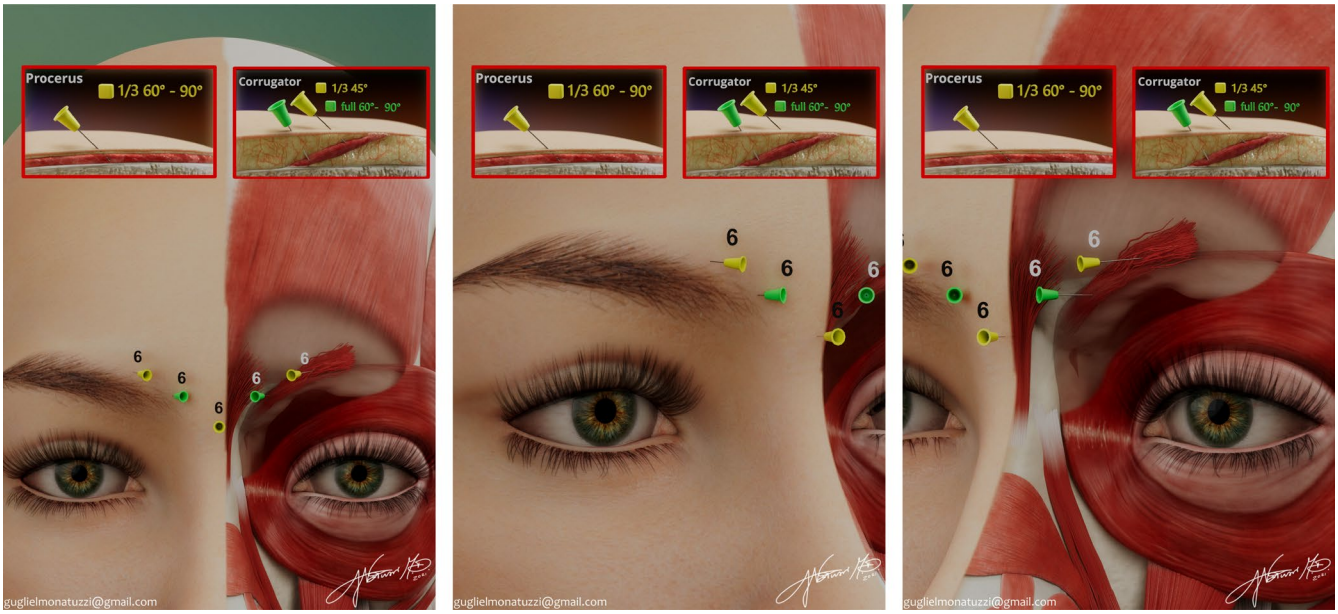


FIGURE 1 Treatment of glabellar lines. The usual total dose is 30 U (although it may be increased up to 50 U). The needle should be inserted in a superolateral direction when injecting the corrugators



FIGURE 2 Treatment of crow's feet lines. The total dose is 18 U per side with injections given in a medial-to-lateral direction. If the orbicularis oculi muscle is much stronger than usual or the lines continue far laterally, an extra 2 U may be injected into two further sites on each side of the face, located laterally to the standard pattern (not shown here)—thus giving a total dose of 22 U per side

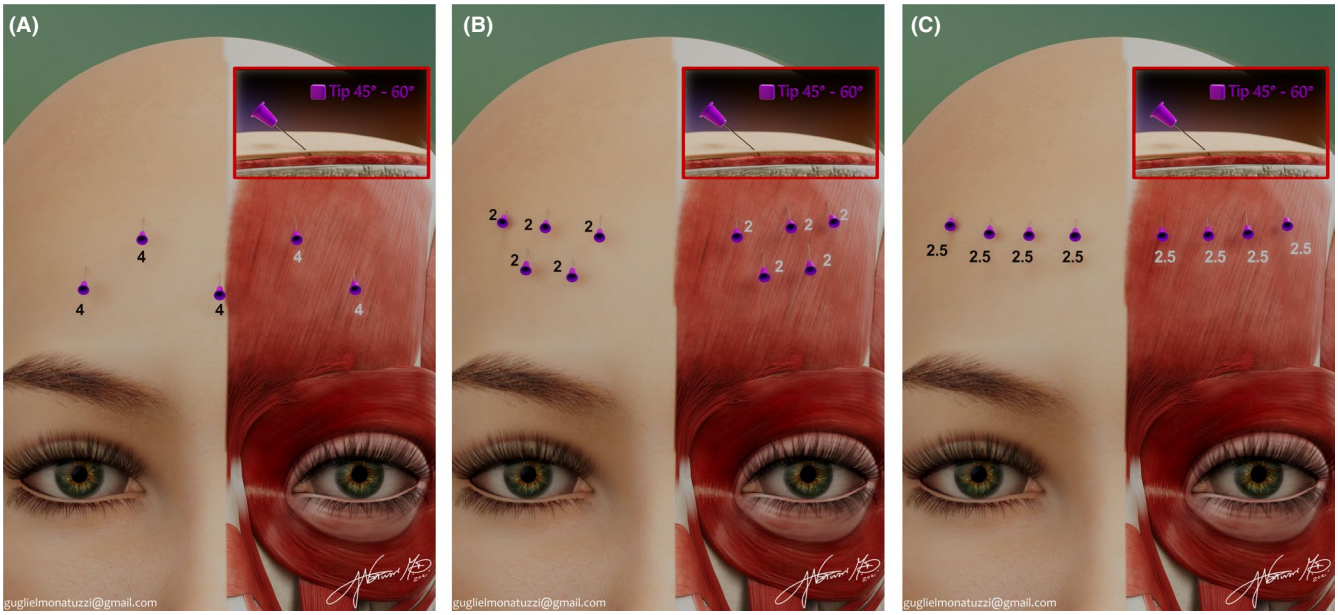


FIGURE 3 Treatment of forehead lines. The total dose is 20 U. Injections may be given using the standard pattern described in the product SmPC (part A). Alternatively, to better distribute the toxin, it can be injected into five sites per side on two levels (in patients with a high forehead, part B) or four sites per side on a single level (in patients with a low forehead; part C)

risk of bruising, which is important for esthetic reasons but also has safety implications because it reduces the likelihood of toxin diffusion into the orbit, which can cause diplopia. In addition, injections should not be performed beyond the peak of zygomatic bone protrusion because this may affect the zygomatic muscle, thereby altering smile dynamics.

Patients with a history of lower eyelid swelling should not be injected in this area because the toxin can decrease muscle tone and exacerbate the issue. Other contraindications include significant laxity of the lower eyelids (identified using the snap test), scleral show, and ectropion.

3.6 | Forehead lines

The interplay of the levator muscle of the forehead (frontalis) with various depressors (procerus, corrugator and orbicularis) is fundamental to the dynamic and static balance of the upper face. Forehead lines are caused by the activity of the frontalis muscles in raising the eyebrows.

Superficial injection is highly recommended by the panel. Needle penetration should not exceed one third and should be made at a 45–60 angle. As with crow's feet treatment, superficial toxin deposition in the deep dermis above the muscle may reduce the risk of

bleeding. Furthermore, erroneous injection deep into the laxer preperiosteal plane can lead to gravitational descent into the orbital area leading to unwanted effects on the eyelid levator muscle.

The total recommended dose of onabotulinumtoxinA is 20 U across both sides of the forehead in females and males. The clinical studies that led to the approval of this dose for forehead lines showed that it was effective, safe and long-lasting, with good preservation of eyebrow position.[23-26] Higher doses may increase the risk of eyebrow ptosis and/or a frozen forehead, leading to a reduced range of expression. The injection pattern described in the product SmPC is shown in Figure 3A. In addition, to better distribute the toxin, the panel recommends injecting into four or five locations bilaterally—using five sites on two levels for patients with a high forehead (Figure 3B) or four sites on a single level for those with a low forehead (Figure 3C).

Ptosis of the eyebrows is the most serious complication that can result from this treatment. It occurs much more frequently than eyelid ptosis, although the esthetic and functional implications are typically less severe. The risk can be decreased by staying ≥2 cm above brow height while injecting and by not exceeding the recommended doses. Nonetheless, injection of the frontalis muscle will always have some effect on eyebrow position, and this should be taken into account when planning the treatment. In general, low doses minimize the risk, but may also be associated with a reduced duration of effect.

Another potential unwanted outcome of forehead treatment is the so-called "Mephisto" or "Spock" eyebrow. This can occur when only the medial portion of the frontalis muscle is treated, and hence onabotulinumtoxinA should always also be administered in the lateral area. The general principle is that the entire muscle must be injected to avoid compensatory overaction of the untreated portion.

Forehead treatment may also lead to asymmetries. These are typically mild and can be rectified using very low doses of onabotulinumtoxinA (0.5–1 U) injected where muscle relaxation is needed to restore balance between the two sides. At-rest asymmetries of the eyebrows evident ahead of injection should be highlighted before proceeding—particularly with new patients.

In individuals with high eyebrows on one or both sides and pronounced activity of the frontalis at rest, possible underlying visual obstructions (eg, eyelid ptosis or dermatochalasis) should be examined. These need surgical rather than toxin-based treatment. Indeed, by reducing frontalis overaction, the use of onabotulinumtoxinA would likely only make the problem worse.

3.7 | Brow lifting and shaping

Lifting and shaping of the brow with onabotulinumtoxinA can be achieved through injection of the glabella and the upper lateral portion of the orbicularis oculi—which lower the eyebrows medially and laterally, respectively. Treatment of these depressor muscles facilitates lifting of the entire eyebrow or of the lateral area only if preferred. Injection of the upper section of the lateral orbicularis oculi can be used to simultaneously treat upper crow's feet lines. This procedure is most often requested by women.

Guidance has already been provided on injection sites, doses and techniques for treating glabellar and crow's feet lines (Figures 1 and 2). Enhanced lifting and shaping of the brow arch may be achieved using an additional injection of 2 U of onabotulinumtoxinA near the tail of the eyebrow (Figure 4). The recommended depth is superficial, as per the usual treatment of crow's feet lines. Potential adverse events are similar to those already discussed for glabellar and crow's feet treatments.

3.8 | Orbicularis oculi hypertrophy in the lower eyelid

This hypertrophy typically manifests as a spindle-like bulge along the pretarsal section of the lower eyelid, just below the ciliary margin. Treatment can reduce the degree of unsightly protrusion. Some panel members also inject this area to open up a constricted eyelid aperture. However, it is important to understand that this treatment does not reduce the appearance of lines on the lower lids.

The total dose recommended by the panel is 0.5–1 U per side for female and male patients (Figure 5). The preferred injection site is in the main area of hypertrophy immediately lateral to the midpupillary line—around 2–4 mm below and parallel to the lid margin. Injections should be given intradermally in a lateral-to-medial direction with the needle tip at a 30 angle. Contraindications include significant laxity of the lower eyelid (confirmed by a positive snap test), scleral show, or ectropion.

3.9 | Bunny lines

The muscles responsible for bunny lines are the transverse part of the nasalis and the levator labii superioris alaeque nasi. Treatment involves a single injection per side into the middle of the impacted area of the nose (Figure 6). The recommended dose for both females and males is 2–3 U per site. Injections should be made at 45 to the surface of the nose with a low-to-high direction. The muscle is located just below the skin and hence superficial penetration is recommended, using only the tip of the needle. Treatment must not go too lateral to avoid involving the levator labii superioris alaeque nasi muscle, which may lead to unwanted lengthening of the upper lip.

Bunny lines are not always a primary finding but are sometimes instead a "tell-tale" sign of previous crow's feet treatment. In such cases, the bunny lines are caused by compensatory overaction of the medial part of orbicularis oculi. However, treatment should be as already described.

3.10 | Sagging nasal tip

Hypertonia of the depressor septi nasi muscle can cause lowering of the nasal tip, which is typically more obvious during smiling.

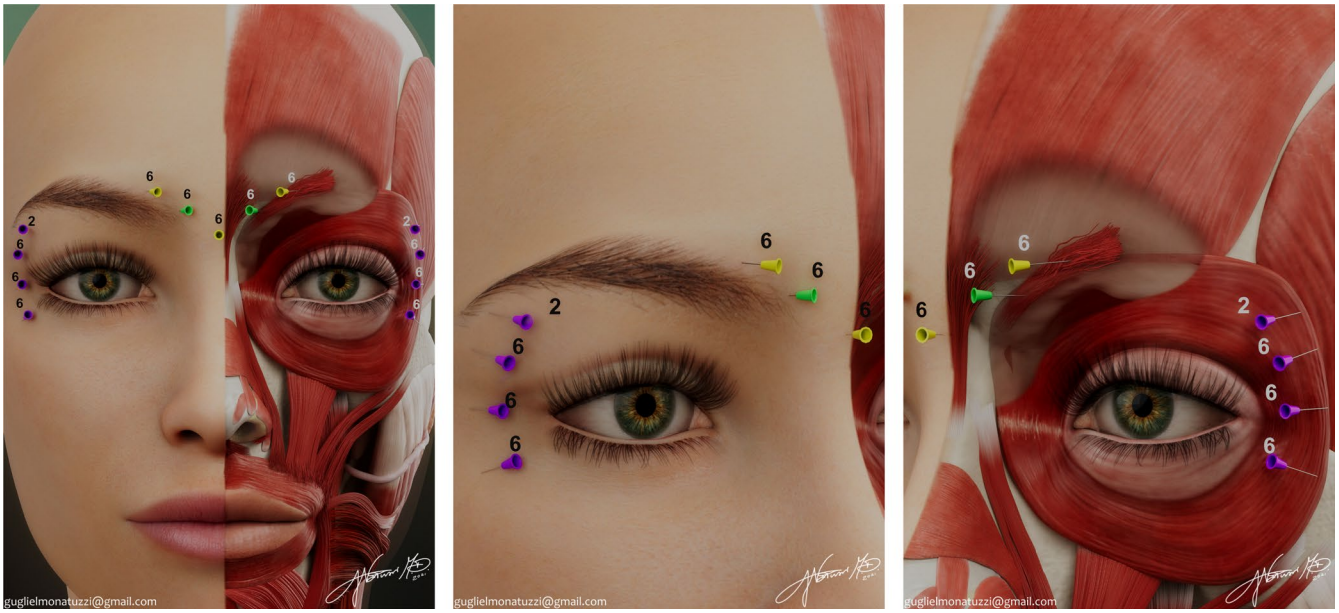


FIGURE 4 Brow lifting and shaping. To lift and shape the brow arch during treatment of the glabella and crow's feet lines, an extra injection can be made into the eyebrow tail area, using a recommended dose of 2 U per side given subcutaneously

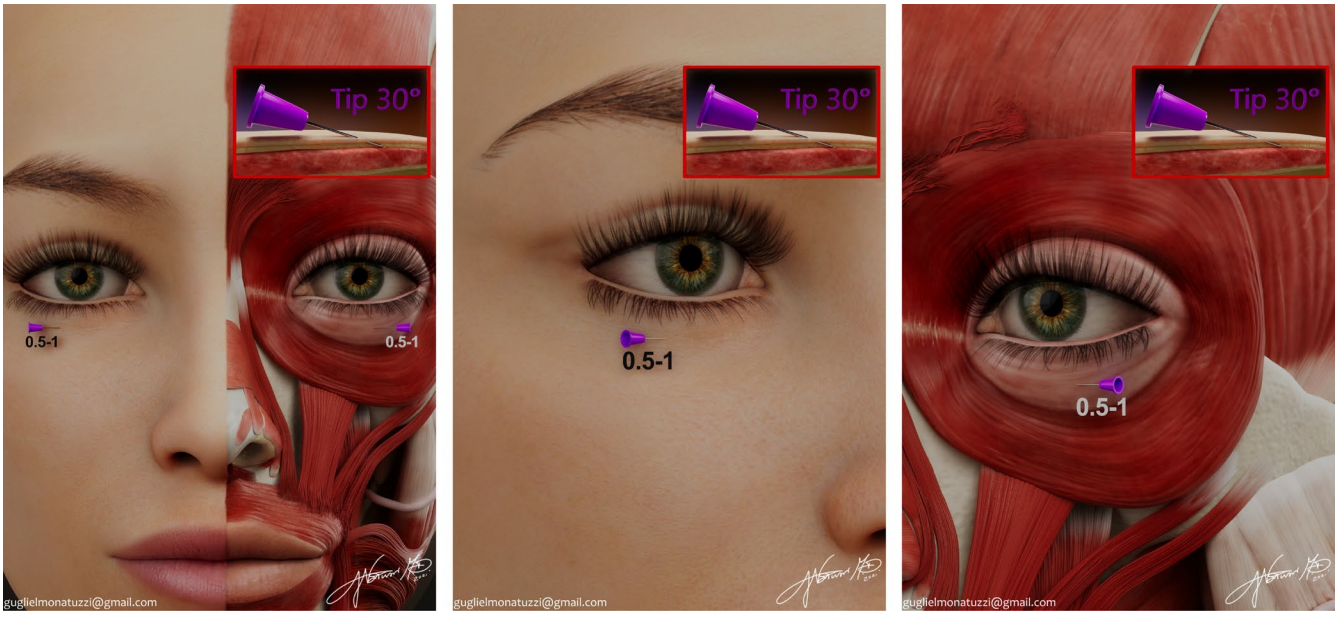


FIGURE 5 Treatment of lower eyelid hypertrophy. The recommended dose is 0.5–1 U per side, with injections given just lateral to the mid-pupillary line at the point of main muscle hypertrophy and performed in a lateral-to-medial direction

Improvement can be achieved using a single injection of 2–4 U of onabotulinumtoxinA in female or male patients (Figure 7). The procedure should be undertaken through full insertion of the needle into the base of the columella, directed at the nasal spine. As a result, the nasal tip is typically rotated in an upward direction, and downward rotation during speaking or smiling is reduced. However, facial anatomy should be carefully assessed before initiating treatment; key contraindications include the presence of a long upper lip or of a long caudal septum, which may reduce upward rotation of the nasal tip.

3.11 | Gummy smile

The level of consensus within the panel was only 60% (6/10 members), and hence this section is considered to be a "position statement."

For correction of a gummy smile, the panel recommends treatment at three sites using 2 U of onabotulinumtoxinA at each—with bilateral injections targeting the levator labii superioris alaeque nasi and a single central injection at the base of the columella targeting the depressor septi nasi (Figure 8). The two lateral treatments should

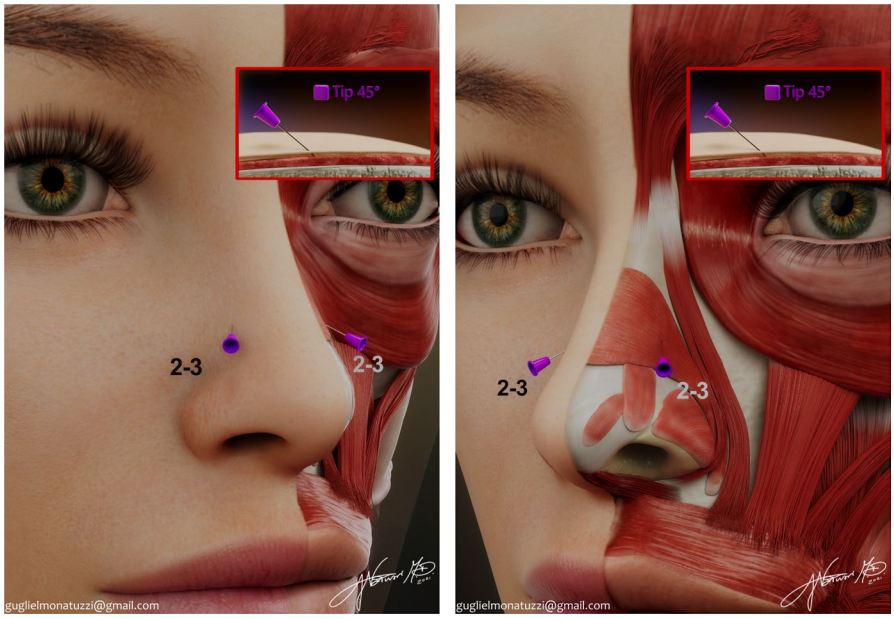


FIGURE 6 Treatment of bunny lines. The total dose is 2–3 U per side based on a single injection given in a low-to-high direction

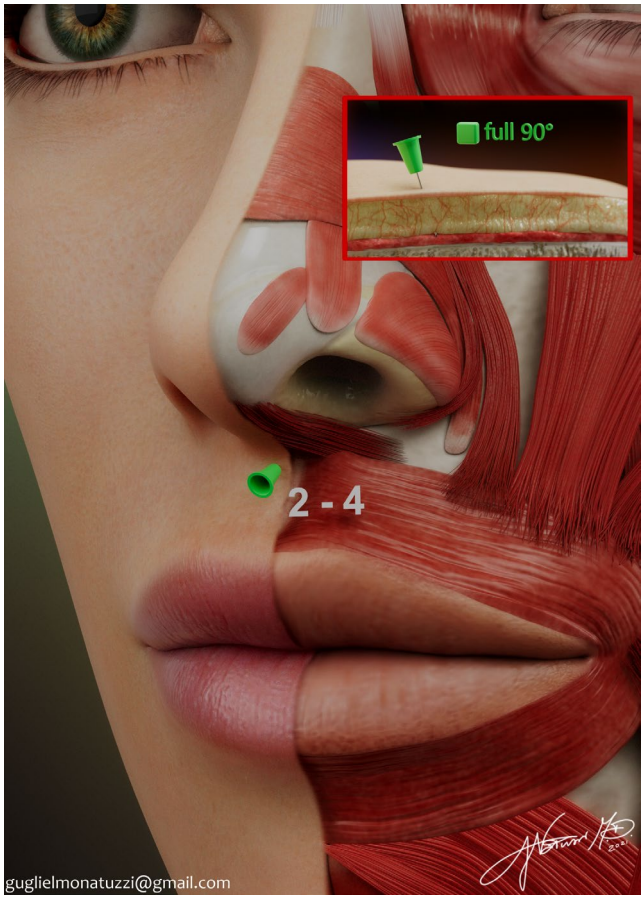


FIGURE 7 Treatment of sagging nasal tip. The total dose is 2–4 U given as a single injection undertaken through full needle insertion at the base of the columella, directed at the nasal spine

be made based on half needle insertion at a site located 1 cm above the base of the nostril and 3–4 mm lateral to it, with the aim of reducing levator labii superioris alaeque nasi function without affecting the orbicularis oris. For the central site, injection can proceed as already described for a sagging nasal tip.

Patients should return for evaluation 2 weeks after the initial treatment. If their gummy smile is under-corrected, further treatment is recommended, targeting the levator labii superioris. Two injections should be made—one per side—into points located 1 cm laterally and below the previous treatment sites (Figure 8). Again, the recommended dose is 2 U per side with half-length needle penetration.

Several recent studies have demonstrated that BoNTA is safe and effective in the treatment of gummy smile.[27-29] Nonetheless, we must stress that it should only be attempted by expert practitioners. Imprecise injection site selection or dosing can result in significant asymmetries of the smile. This method is also contraindicated in patients who already have a long upper lip because treatment can make it appear even longer. Moreover, BoNTA-based treatment is not recommended for individuals with a severe defect (ie, those in whom gum tissue display is 50%–100% of teeth length). Such patients may be deemed to have gingival hypertrophy, requiring surgical correction.[30,31]

Finally, the introduction by de Maio of the concept of "myomodulation" with hyaluronic acid fillers has given rise to a new alternative in the treatment of gummy smile.[32,33] This may be particularly indicated in patients with a long upper lip and is also longer lasting.

3.12 | Masseter hypertrophy

Treatment of masseter hypertrophy with onabotulinumtoxinA may be undertaken for esthetic purposes (to reduce unsightly muscle mass at the mandibular angle) and/or for function improvement of conditions such as bruxism. The anterior and posterior limits of the muscle can be easily demarcated by palpating the area while the patient clenches their teeth; the upper margin will be at the

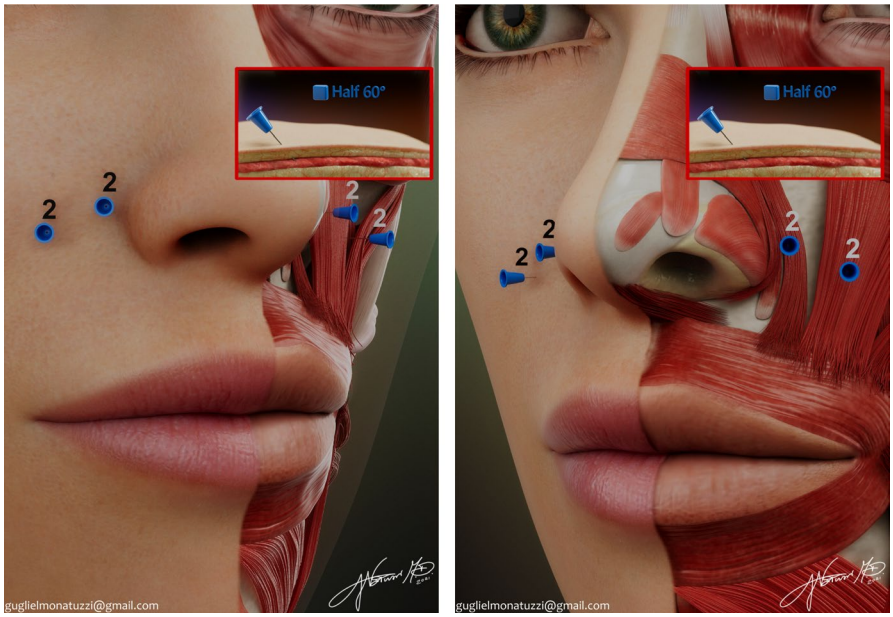


FIGURE 8 Treatment of gummy smile. The initial treatment is based on three sites using 2 U in each: bilateral injections targeting the levator labii superioris alaeque nasi (1 cm above the base of the nostril and 3–4 mm lateral to it) and a single central site at the base of the columella targeting the depressor of the nasal septum (not shown here but proceeding as per correction of a sagging nasal tip). If the gummy smile is under-corrected, further injections can be given using 2 U per side (1 cm lateral and below the previous treatment sites) targeting the levator labii superioris

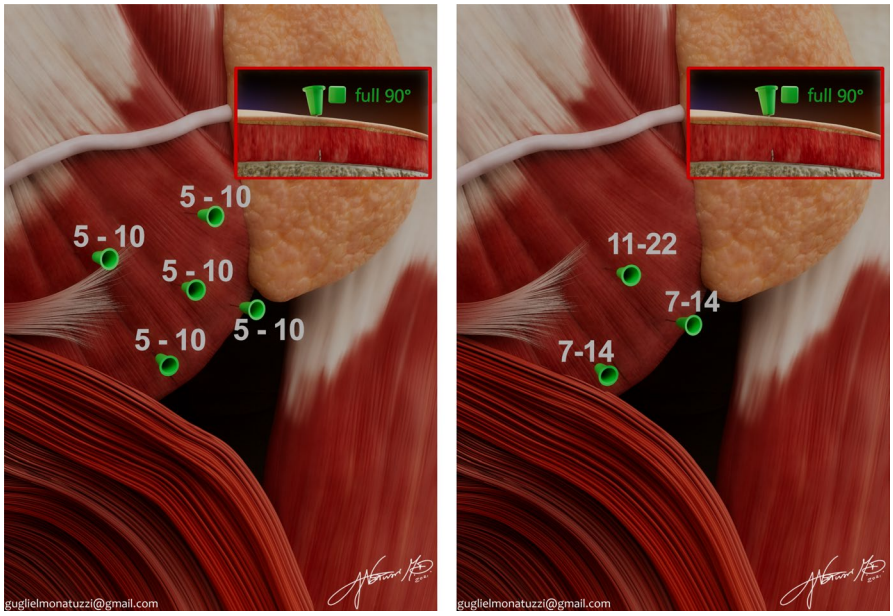


FIGURE 9 Treatment of masseter hypertrophy. The total dose is 25–50 U per side (according to muscle mass and strength) using between three and five injection sites. Possible injection patterns are shown based on five injection sites (part A) or three sites (part B)

zygomatic arch and the lower margin will be along the inferior mandibular edge.

The panel recommends using between three and five injection sites according to muscle area, and the total onabotulinumtoxinA dose may range from 25 to 50 U per side in females and males depending on muscle strength and mass (Figure 9). The recommended device is a 30G X 1" needle inserted perpendicularly all the way to the bone and then retracted by 1–2 mm before commencing the injection. To ensure an even distribution across the deep and superficial masseter, the bolus should be inserted while slowly pulling back the needle. A one-inch needle is required to reach all the way down and reduces the potential for erroneously injecting a hypertrophic parotid gland. Injection of the full thickness of the masseter is important to avoid herniation of untreated portions of the muscle with a very visible bulging of the soft tissues at the mandibular angle.[34]

The onset of effects takes substantially longer when treating the masseter compared with other applications because the aim of treatment is different. Whereas wrinkle reduction and modulation of expression will be evident as soon as chemodenervation has taken place, treatment of muscle hypertrophy requires many weeks of decreased activity in order for relative atrophy to occur.[35] Hence, injectors should wait at least 4–6 weeks before deciding whether further onabotulinumtoxinA treatment is needed—with the number of units determined based on the overall clinical assessment. Furthermore, the interval between treatment cycles may vary substantially. Thus, periodic reassessment is required, with the treatment repeated once muscle strength begins to come back rather than waiting for the reappearance of hypertrophy.

A Phase 3 study of onabotulinumtoxinA in patients with a wide lower face due to masseter muscle prominence is now ongoing

(NCT04073303) and should shed further light on optimal practice for this treatment.

3.13 | Perioral lines

The prevention or treatment of perioral lines typically requires a combination approach based on onabotulinumtoxinA and a hyaluronic acid (HA) filler—with the filler following 7–10 days after BoNTA treatment. In this way, targeted injection of onabotulinumtoxinA can be used to weaken the outer fibers of the orbicularis oris muscle, decreasing the quantity of filler needed to achieve a good result without unduly affecting the volume and shape of the lips.

Treatment can proceed based on one injection site on each side of the mouth, located 1 mm above the vermilion border of the upper lip (or below the lower lip) and around 5 mm from the apex of the Cupid's bow. If required, two further injection sites can be added—one per side per lip—inserting the needle 1 mm above the vermilion border and at least 5 mm medial to the oral commissure (Figure 10). Injections should be superficial, with the needle almost parallel to the surface of the skin, the bevel facing upwards, and only the tip inserted. Whether injecting two or four locations, the preferred dose is 0.5–1 U per site, based on clinical judgment.

Proprioceptive disorders are possible even with optimal technique, and some patients describe feeling an abnormal sensation when moving their lips—although mobility may appear outwardly normal. Prior to injection, they should be informed that such disorders typically occur only after the first injection cycle and usually last no longer than 15–20 days without altering overall articulation. However, excessive doses can cause asymmetry and functional lip disorders, and injectors are strongly advised not to exceed the doses recommended here.

3.14 | Marionette lines and "sad mouth"

The primary cause of marionette lines is soft tissue ptosis in the cheeks, although hypertonia of the depressor anguli oris muscle can also be partly responsible. This can lead to depression of the oral commissures, giving a sad look to the patient's face.

OnabotulinumtoxinA treatment should be performed using a single site on each side of the face, injecting 1 cm laterally of the labiomental fold and 1 cm above the lower mandibular margin (Figure 11). Only one third of the needle should be inserted, at an angle of 90 to the skin surface. The recommended dose is 2–4 U per side for both females and males. Injection of this area risks affecting the depressor labii inferioris muscle located just below the depressor anguli oris, and this can lead to unacceptable effects on smile dynamics. In such cases, injection of the opposite depressor labii inferioris with 1–2 U of BoNTA may be considered to improve symmetry. Impaired lip function is also possible if the orbicularis oris is reached by the toxin, leading to increased patient discomfort. Superficial injections at this level are key to avoiding complications.

3.15 | Mentalis muscle hypertonia

The mentalis is a strong mimic muscle with fibers ordered in a vertical direction from the bone and expanding in a branch-like manner toward the skin—a pattern that is quite distinct from other mimic muscles. It functions as an elevator of the chin soft tissue and of the lower lip. However, hypertonia of the mentalis can lead to flattening of the chin against the underlying bone, and may produce a pebbled appearance of the skin (sometimes referred to as a "golf-ball" or "peau d'orange" look), resulting from the dermal insertions of the muscle. Reducing mentalis activity can improve projection of the soft tissue and reduce pebbling of the skin.

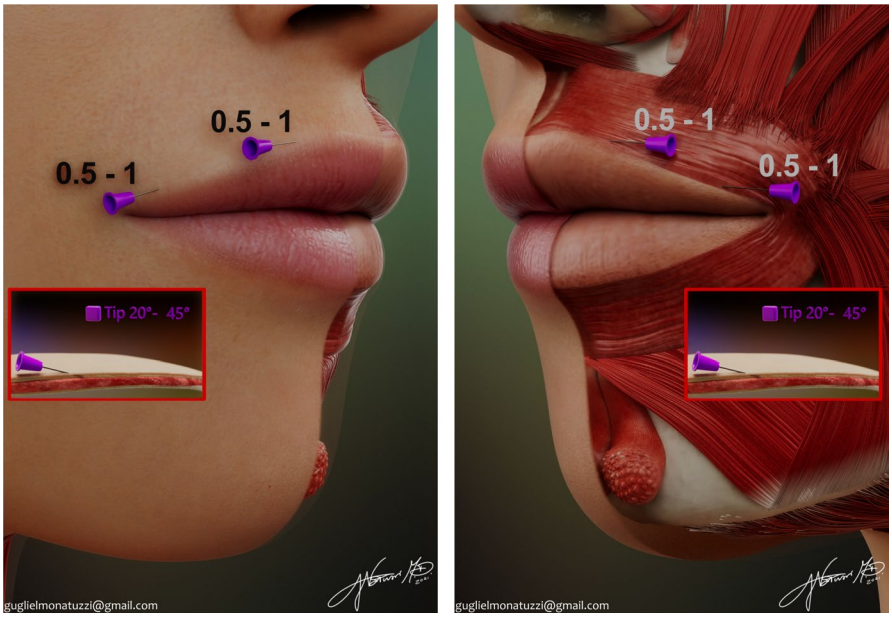


FIGURE 10 Treatment of perioral lines. Treatment can be performed using 0.5–1 U at one injection site per side (1 mm above the vermilion border of the upper lip—or below the lower lip—and around 5 mm from the apex of the Cupid's bow). If required, one further injection sites can be added on each side of the lip, 1 mm above the vermilion border and at least 5 mm medial to the oral commissure. The needle should be almost parallel to the skin surface with the bevel facing upwards, and penetration should be superficial using the tip only

The total recommended dose is 8–10 U in both female and male patients. Two different treatment patterns can be used depending on chin width—based on either one injection site in the midline of the chin (Figure 12A,B) or bilateral sites located 5 mm from the midline (Figure 12C,D). With either pattern, injections should be 1 cm above the lower margin of the chin with needle insertion of half-to-full depth.

When a bilateral technique is used, the distance of 5 mm from the midline must not be exceeded to avoid undesired effects on the depressor labii inferioris muscle, which can result in asymmetry of the lower lip. If this does occur, the opposite depressor labii inferioris may be injected with 1–2 U of onabotulinumtoxinA to correct the asymmetry. Similarly, whether using one or two locations, these sites should be no more than 1 cm from the lower margin of the chin to avoid involving the orbicularis oris and hence causing unwanted effects on lip function.

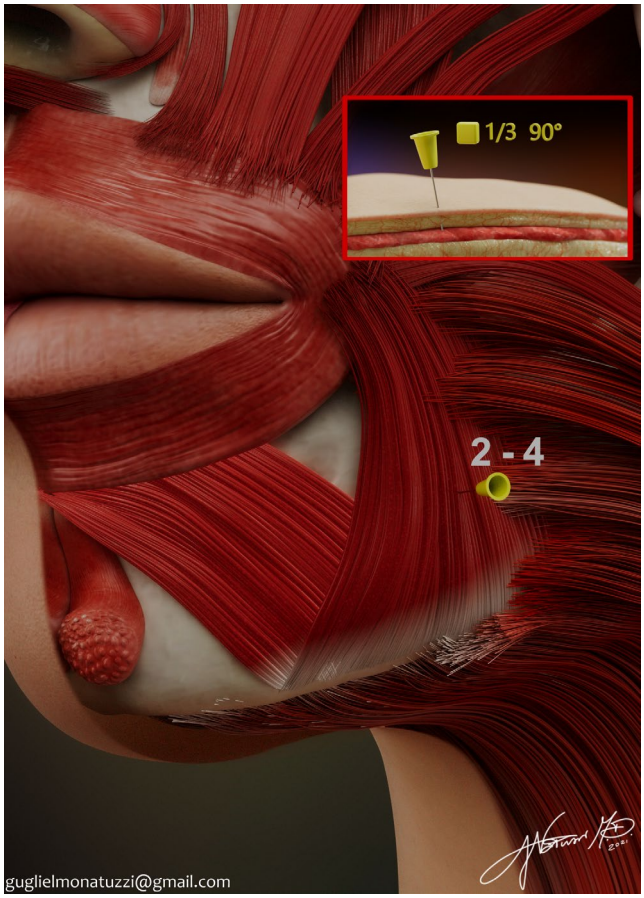


FIGURE 11 Treatment of marionette lines. The total dose is 2–4 U per side using an injection site located 1 cm laterally of the labiomental fold and 1 cm above the lower mandibular margin.

3.16 | Hypertonic platysma bands

Platysma bands typically present in patients older than 45–50 years and show various patterns—single or multiple, medial and/or lateral. In appropriately selected patients, good improvements can often be obtained through treatment with onabotulinumtoxinA, particularly in individuals that are relatively young and have elastic skin without significant gravity-related sagging.

The injection pattern should be adapted to the location of the bands: medial (most common), lateral (very uncommon), or mixed (relatively common). The patient should be assessed during dynamic activity, such as smiling, talking, and teeth clenching.

The panel recommends Injecting at sites positioned 2 cm apart along each platysma band, in most cases using four locations per band (Figure 13). The preferred dose is 2 U per injection, thus yielding an overall dose of 6–10 U for each band; all bands, whether medial or lateral, can typically be injected in the same session. The needle insertion depth should be very superficial (tip only), delivered almost parallel to the skin. This is a particularly crucial consideration because the anterior muscles of the neck lie below the thin platysma muscle and must never be impacted by the BoNTA injection if severe side effects are to be prevented.

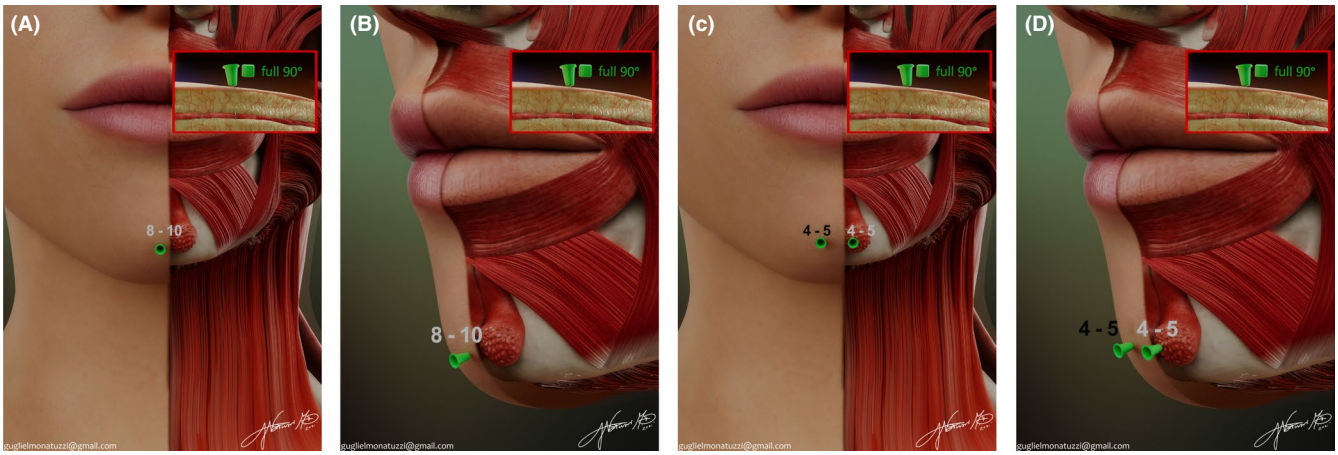


FIGURE 12 Treatment of mentalis hypertonia. The total recommended dose is 8–10 U. Two different treatment patterns can be used depending on chin width—based on either one injection site in the midline of the chin (parts A and B) or bilateral sites located 5 mm from the midline (parts C and D). With either pattern, injections should be 1 cm above the lower margin of the chin

3.17 | Nefertiti lift

The "Nefertiti lift" is the attractive name given to a technique first described by Levy in 2007.[36] The goal is to redefine the jawline by counteracting the down-pull of the platysma. The initially proposed technique was based on a single line of injections performed along the mandibular margin, as well as in the upper part of the posterior platysma band, using 15–20 U of onabotulinumtoxinA on each side of the face.[36] This was an important innovation but perhaps ahead of its time, and it did not gain widespread popularity at first. Subsequently, de Maio[37] recommended simultaneous treatment of the platysma along the mandibular margin in two parallel rows using 10–20 U per side, the posterior band using 8–10 U per side, and the anterior band using 6–9 U per side (de Maio M, personal communication). This concept was then modified by de Almeida to include two parallel rows of injections above and below the mandibular margin along with treatment of the mentalis, using a total of 16–20 U per side.[38]

Owing to these modifications—and growing interest in treating the lower face and neck with BoNTA—variations of the Nefertiti lift are becoming increasingly popular. At the doses recommended above and based on the technical specifications provided in the relevant references, these procedures can be effective and safe. A superficial injection technique is essential.

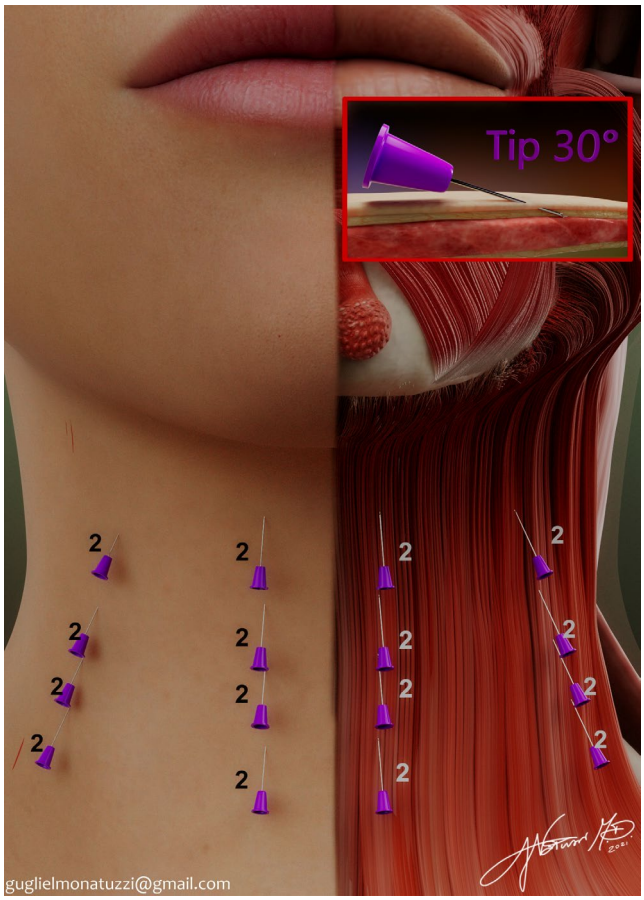


FIGURE 13 Treatment of hypertonic platysma bands. The total dose is 6–10 U for each platysma band (2 U per site). Injection points should be located every 2 cm along each one, totaling four per band in most cases. The needle insertion depth should be very superficial, delivered almost parallel to the skin

3.18 | Microbotox

The concept of "Microbotox" is based on injecting microdroplets of a diluted solution of onabotulinumtoxinA into the deep dermis of selected areas of the face and neck. The aim is to improve skin quality through action on the dermal adnexa (sebaceous and sweat glands and arrector pili muscles), while still obtaining a mild effect on the deeper mimic muscles through diffusion. It was first introduced by Wu in 2002 (initially termed "Mesobotox"),[39] and was later expanded upon and further popularized by the same author in 2015.[40]

The recommended onabotulinumtoxinA concentration is 50% of that used for standard neuromodulation—thus 2.5 ml for a 50 U vial or 5 ml for a 100 U vial. Pure saline may be used, or saline plus lidocaine as proposed by Wu.[40] The technique is key: only microdroplets should be injected, strictly intradermally, at a distance of about 1 cm between each point.

According to individual clinical features, the upper, mid and lower face and/or neck can be treated, but appropriate patient selection is essential for a good result. In particular, individuals with significant skin relaxation should not be treated with this technique.

4 | DISCUSSION

Patients seeking esthetic improvement of the face and neck now have more options than ever before. Despite this, BoNTA remains comfortably the most frequently used treatment modality worldwide.[1,2] Given its enduring popularity, as well as the wide variety of different specialties that can administer treatment (plastic surgeons, dermatologists, esthetic physicians, etc.), up-to-date technical guidance is crucial. This updated consensus report was developed to support improved standardization of onabotulinumtoxinA procedures, and to optimize safety and effectiveness, based on our experiences in Italy as well as reports from other countries.

4.1 | Importance of patient assessment

Careful patient assessment is crucial before initiating any treatment. In most cases, a customized injection plan will be required to achieve the desired results, with both general and more specific patient characteristics being considered. Among general parameters, information on sex, age, ethnicity, skin type, and history should be collected. The more specific parameters assessed will depend on the area(s) being treated. For example, in the forehead, key characteristics may include overall size, wrinkle distribution, and eyebrow position. These assessments should be performed both statically and dynamically. Furthermore, although the target tissue with onabotulinumtoxinA is typically muscular, careful evaluation should be made

of other structures that influence aging, including the skeleton, skin, and vascularization.

In recent years, men have become an increasingly important demographic for esthetic medicine in general—and BoNTA in particular.[2,11] Anatomical differences between genders are, therefore, a key consideration.

Irrespective of patient characteristics, photographic documentation is essential for the analysis of results, assessment of procedures, and implementation of any changes needed for greater personalization of treatment. Photographic records also facilitate the creation of historical archives that can serve valuable scientific and clinical functions.

4.2 | Post-treatment recommendations

In the early days of esthetic treatment with BoNTA, patients were typically advised to limit physical exercise, and avoid make-up, any form of pressure on the face, sauna, etc., for up to 7 days post-treatment. It was feared that these activities could cause spreading of the toxin to surrounding areas with consequent complications. Over the years, this period of caution has been gradually reduced, but many physicians still recommend such limitations for a day or two.

However, these restrictive instructions should now be discontinued—they are not supported by scientific evidence and can generate in patients the impression of a potentially dangerous treatment. Indeed, it has been clarified that most BoNTA is inside the synaptic vesicles in muscles within 5–10 min after binding.[41]

4.3 | Managing patient expectations

Information on the uses, effectiveness and safety of esthetic treatment with onabotulinumtoxinA is freely available to all patients on the internet—at any time and independently of clinician control. The quality of these resources varies widely.[42] This can cause an overload of potentially incorrect or contradictory information and may lead to unrealistic expectations. All clinicians should, therefore, see it as an obligation to guide patients through the process by delivering a complete package of material on the potential indications, outcomes and risks of BoNTA treatment. Based on this, patients should then be free to make an informed decision on whether or not to proceed. Photographs can also be a valuable tool for communicating the therapeutic plan and illustrating the results. As with esthetic plastic surgery, it is essential that patients sign an adequate informed consent form.[43]

5 | CONCLUSIONS

Demand for esthetic treatment of the face and neck using BoNTA continues to grow worldwide. The present paper provides consensus-based recommendations on the use of onabotulinumtoxinA for this purpose, developed by a panel of 10 highly experienced practitioners from various specialties. Standardized techniques are a key element of optimal practice, in order to maximize safety and effectiveness across the full range of potential uses.

ACKNOWLEDGEMENTS

The authors thank Dr Guglielmo Natuzzi for the development of anatomical figures. These were funded by Allergan SpA with an unconditional grant. Writing and editorial assistance was provided to the authors by Timothy Ryder, DPhil, of Biological Communications Limited (London, United Kingdom) and funded by Allergan SpA at the request of the investigator. All authors met the ICMJE authorship criteria. Neither honoraria nor payments were made for authorship.

CONFLICT OF INTEREST

All of the authors are consultants and speakers for Allergan Aesthetics. None has any financial interest in the products mentioned in this article.

AUTHOR CONTRIBUTIONS

All authors contributed to the conception and design of the paper and to the creation of consensus; all were involved in drafting the manuscript and revising it critically for important intellectual content; and all gave final approval of the version to be published.

ETHICAL APPROVAL

The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered to. No ethical approval was required as this article contains no original research data.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current work.

ORCID

Dario Bertossi https://orcid.org/0000-0002-8635-9967

REFERENCES

1. American Society of Plastic Surgeons. National plastic surgery statistics 2020. www.plasticsurgery.org/documents/News/Statistics/2020/plastic-surgery-statistics-report-2020.pdf. Accessed September 2, 2021.
2. International Society of Aesthetic Plastic Surgery. ISAPS international survey on aesthetic/cosmetic procedures performed in 2019. www.isaps.org/wp-content/uploads/2020/12/Global-Survey-2019.pdf. Accessed September 2, 2021.
3. Bertossi D, Cavallini M, Cirillo P, et al. Italian consensus report on the esthetic use of onabotulinumtoxinA. *Esperienze Dermatologiche*. 2016;18:45-57.
4. Bertossi D, Cavallini M, Cirillo P, et al. Italian consensus report on the aesthetic use of onabotulinum toxin A. *J Cosmet Dermatol*. 2018;17:719-730.

5. Vistabel. Summary of product characteristics. www.medicines.ie/medicines/Vistabel-34207/spc. Accessed September 2, 2021.
6. Cox SE, Joseph JH, Fagien S, et al. Safety, pharmacodynamic response, and treatment satisfaction with onabotulinumtoxinA 40 U, 60 U, and 80 U in subjects with moderate to severe dynamic glabellar lines. Paper presented at: *2020 American Society for Dermatologic Surgery Annual Meeting*; October 9–11, 2020; virtual meeting.
7. D'Emilio R, Rosati G. Full-face treatment with onabotulinumtoxinA: results from a single-center study. *J Cosmet Dermatol*. 2020;19:809-816.
8. Werner M. *Manual of Aesthetic Surgery*. Springer-Verlag Berlin Heidelberg; 2011.
9. Keen JA. A study of the differences between male and female skulls. *Am J Phys Anthropol*. 1950;8:65-79.
10. Lorenc ZP, Kenkel JM, Fagien S, et al. Consensus panel's assessment and recommendations on the use of 3 botulinum toxin type A products in facial aesthetics. *Aesthet Surg J*. 2013;33(Suppl):35S-40S.
11. Keaney TC, Anolik R, Braz A, et al. The male aesthetic patient: facial anatomy, concepts of attractiveness, and treatment patterns. *J Drugs Dermatol*. 2018;17:19-28.
12. Al-Hoqail RA, Meguid EM. Anatomic dissection of the arterial supply of the lips: an anatomical and analytical approach. *J Craniofac Surg*. 2008;19:785-794.
13. Carruthers J, Fournier N, Kerscher M, et al. The convergence of medicine and neurotoxins: a focus on botulinum toxin type A and its application in aesthetic medicine – a global, evidence-based botulinum toxin consensus education initiative: part II: incorporating botulinum toxin into aesthetic clinical practice. *Dermatol Surg*. 2013;39:510-525.
14. Allen SB, Goldenberg NA. Pain difference associated with injection of abobotulinumtoxinA reconstituted with preserved saline and preservative-free saline: a prospective, randomized, side-by-side, double-blind study. *Dermatol Surg*. 2012;38:867-870.
15. Zidan A, Hussaini S, Gibson S, Brooks G, Mejico L. Onabotulinumtoxin type A reconstitution with preserved versus preservative-free saline in chronic migraine (B-RECON). A randomised, double-blind trial. *Int J Clin Pract*. 2020;74:e13522.
16. Güleç AT. Dilution of botulinum toxin A in lidocaine vs. in normal saline for the treatment of primary axillary hyperhidrosis: a double-blind, randomized, comparative preliminary study. *Eur Acad Dermatol Venereol*. 2012;26:314-318.
17. Kim A, Jung J, Pak A. Botulinum toxin type A reconstituted in lidocaine with epinephrine for facial rejuvenation: results of a participant satisfaction survey. *Cutis*. 2013;Suppl:13-18.
18. Sezgin B, Ozel B, Bulam H, Guney K, Tuncer S, Cenetoglu S. The effect of microneedle thickness on pain during minimally invasive facial procedures: a clinical study. *Aesthet Surg J*. 2014;34:757-765.
19. Bertossi D, Mortellaro C, Nocini P. New clinical analysis and device for Botox injections. *J Craniofac Surg*. 2016;27:1554-1557.
20. Park MY, Ahn KY. Effect of the refrigerator storage time on the potency of Botox for human extensor digitorum brevis muscle paralysis. *J Clin Neurol*. 2013;9:157-164.
21. Vagin O, Tokhtaeva E, Garay PE, et al. Recruitment of septin cytoskeletal proteins by botulinum toxin A protease determines its remarkable stability. *J Cell Sci*. 2014;127:3294-3308.
22. de Maio M, Swift A, Signorini M, et al. Facial assessment and injection guide for botulinum toxin and injectable hyaluronic acid fillers: focus on the upper face. *Plast Reconstr Surg*. 2017;140:265e-276e.
23. Solish N, Rivers JK, Humphrey S, et al. Efficacy and safety of onabotulinumtoxinA treatment of forehead lines: a multicenter, randomized, dose-ranging controlled trial. *Dermatol Surg*. 2016;42:410-419.
24. Fagien S, Cohen JL, Coleman W, et al. Forehead line treatment with onabotulinumtoxinA in subjects with forehead and glabellar facial rhytids: a phase 3 study. *Dermatol Surg*. 2017;43:S274-S284.
25. De Boulle K, Werschler WP, Gold MH, et al. Phase 3 study of onabotulinumtoxinA distributed between frontalis, glabellar complex, and lateral canthal areas for treatment of upper facial lines. *Dermatol Surg*. 2018;44:1437-1448.
26. De Boulle K, Carruthers A, Solish N, et al. OnabotulinumtoxinA treatment for moderate to severe forehead lines: a review. *Plast Reconstr Surg Glob Open*. 2020;8:e2669.
27. Cengiz AF, Goymen M, Akcali C. Efficacy of botulinum toxin for treating a gummy smile. *Am J Orthod Dentofacial Orthop*. 2020;158:50-58.
28. Gong X, Huang H, Gu C, et al. Individual factors of botulinum type A in treatment of gummy smile: a prospective study. *Aesthet Surg J*. 2021;41:NP842-NP850.
29. Rajagopal A, Goyal M, Shukla S, Mittal N. To evaluate the effect and longevity of Botulinum toxin type A (Botox®) in the management of gummy smile - A longitudinal study up to 4 years follow-up. *J Oral Biol Craniofac Res*. 2021;11:219-224.
30. Nahm KY, Shin SY, Ahn HW, Kim SH, Nelson G. Gummy smile correction using lingual orthodontics and augmented corticotomy in extremely thin alveolar housing. *J Craniofac Surg*. 2017;28:e599-e603.
31. Dym H, Pierre R. Diagnosis and treatment approaches to a "gummy smile". *Dent Clin North Am*. 2020;64:341-349.
32. de Maio M. Myomodulation with injectable fillers: an innovative approach to addressing facial muscle movement. *Aesthetic Plast Surg*. 2018;42:798-814.
33. de Maio M. Myomodulation with injectable fillers: an innovative approach to addressing facial muscle movement. *Aesthetic Plast Surg*. 2020;44:1300-1316.
34. Lee HJ, Kang IW, Seo KK, et al. The anatomical basis of paradoxical masseteric bulging after botulinum neurotoxin type A injection. *Toxins*. 2016;9:14.
35. Shome D, Khare S, Kapoor R. Efficacy of botulinum toxin in treating Asian Indian patients with masseter hypertrophy: a 4-year follow-up study. *Plast Reconstr Surg*. 2019;144:390e-396e.
36. Levy PM. The 'Nefertiti lift': a new technique for specific recontouring of the jawline. *J Cosmet Laser Ther*. 2007;9:249-252.
37. de Maio M, Wu WTL, Goodman GJ, et al. Facial assessment and injection guide for botulinum toxin and injectable hyaluronic acid fillers: focus on the lower face. *Plast Reconstr Surg*. 2017;140:393e-404e.
38. de Almeida ART, Romiti A, Carruthers JDA. The facial platysma and its underappreciated role in lower face dynamics and contour. *Dermatol Surg*. 2017;43:1042-1049.
39. Wu WTL. Facial rejuvenation without facelifts—personal strategies. Presented at: *Regional Conference in Dermotological Laser and Facial Cosmetic Surgery 2002*, Hong Kong; September 13–15, 2002.
40. Wu WTL. Microbotox of the lower face and neck: evolution of a personal technique and its clinical effects. *Plast Reconstr Surg*. 2015;136(Suppl):92S-100S.
41. Colasante C, Rossetto O, Morbiato L, Pirazzini M, Molgó J, Montecucco C. Botulinum neurotoxin type A is internalized and translocated from small synaptic vesicles at the neuromuscular junction. *Mol Neurobiol*. 2013;48:120-127.
42. Barbarite E, Shaye D, Oyer S, Lee LN. Quality assessment of online patient information for cosmetic botulinum toxin. *Aesthet Surg J*. 2020;40:NP636-NP642.
43. Nejadsarvari N, Ebrahimi A. Different aspects of informed consent in aesthetic surgeries. *World J Plast Surg*. 2014;3:81-86.

How to cite this article: Signorini M, Piero Fundarò S, Bertossi D, et al. OnabotulinumtoxinA from lines to facial reshaping: A new Italian consensus report. *J Cosmet Dermatol*. 2022;21:550–563. doi:10.1111/jocd.14728